Writing Expressions and Equations

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS expression *(kīanga taurangi)* · equation *(whārite)* · variable *(taurangi)* · sum *(tapeke)*

A Expressions from words ●○○

Remember: An expression has no equals sign. Write multiplication without the sign ($4n$) and division as a fraction ($\frac{n}{2}$).

1. Write an expression for each. Use n for the number.

a) 5 more than a number _____

b) 3 less than a number _____

c) A number multiplied by 4 _____

d) A number divided by 2 _____

e) Double a number, then add 7 _____

f) Add 7 to a number, then double the result _____

g) A number subtracted from 10 _____

h) A number squared _____

i) The product of 6 and a number, minus 5 _____

j) Half of a number, take away 1 _____

k) Three times the sum of a number and 2 _____

l) A number divided by 2, then 5 added _____

B Expressions in context ●●○

2. Write an expression for each.

a) Apples cost \$3 each. Write an expression for the cost, in dollars, of a apples.

b) A taxi charges \$4 plus \$2 for each kilometre. Write an expression for the cost, in dollars, of a trip of k kilometres.

c) Mia is m years old. Her brother is 5 years younger. Write an expression for her brother's age.

d) \$60 is shared equally between p people. Write an expression for the amount each person gets, in dollars.

e) A phone plan costs \$20 a month plus \$2 for each gigabyte of data. Write an expression for the cost, in dollars, of a month using g gigabytes.

f) A rectangle is w cm wide. It is 3 times as long as it is wide. Write an expression for its perimeter.

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C Equations from words ●●○

Remember: An equation has an equals sign. Write the equation first, then solve it.

3. Write an equation for each problem, then solve it to find the number.

a) I think of a number and add 9. The answer is 21.

b) I think of a number, double it and subtract 5. The answer is 13.

c) A number divided by 4 is 7.

d) Three times a number, plus 8, equals 29.

e) When 6 is subtracted from 5 times a number, the answer is 24.

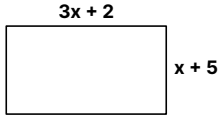
f) Half of a number, plus 3, is 10.

D Expressions for perimeter and area ●●●

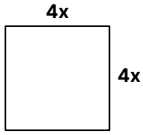
Remember: Perimeter: add all the sides. Area of a rectangle: length × width. Simplify your answer.

4. Write an expression for the perimeter of each shape.

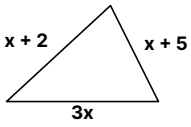
a)



b)

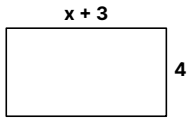


c)

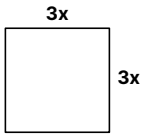


5. Write an expression for the area of each shape.

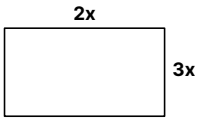
a)



b)



c)



E Solving problems ●●●

6. Write an equation for each problem, then solve it.

a) A rectangle is x cm wide and $(x + 4)$ cm long. Its perimeter is 28 cm. Find its width.

b) An adult ticket costs \$12 and a child ticket costs x dollars. Two adults and 3 children pay \$45 altogether. Find the cost of a child ticket.

c) Ana has x dollars. Ben has twice as much as Ana. Together they have \$51. How much does Ana have?

d) Two consecutive whole numbers add to 41. Find the smaller number.

★ Extension

7. n is a whole number. Write an expression for the sum of n and the next two whole numbers after it. Simplify your answer.

8. A square has sides of s cm. Each side is made 2 cm longer. Write an expression for the perimeter of the new square. Simplify your answer.

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Writing Expressions and Equations

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS expression *(kīanga taurangi)* · equation *(whārite)* · variable *(taurangi)* · sum *(tapeke)*

A Expressions from words ●○○

Remember: An expression has no equals sign. Write multiplication without the sign ($4n$) and division as a fraction ($\frac{n}{2}$).

1. Write an expression for each. Use n for the number.

a) 8 more than a number _____

b) 4 less than a number _____

c) A number multiplied by 6 _____

d) A number divided by 3 _____

e) Double a number, then add 9 _____

f) Add 9 to a number, then double the result _____

g) A number subtracted from 12 _____

h) A number squared _____

i) The product of 5 and a number, minus 7 _____

j) Half of a number, take away 1 _____

k) Four times the sum of a number and 3 _____

l) A number divided by 3, then 8 added _____

B Expressions in context ●●○

2. Write an expression for each.

a) Apples cost \$4 each. Write an expression for the cost, in dollars, of a apples.

b) A taxi charges \$5 plus \$3 for each kilometre. Write an expression for the cost, in dollars, of a trip of k kilometres.

c) Mia is m years old. Her brother is 7 years younger. Write an expression for her brother's age.

d) \$80 is shared equally between p people. Write an expression for the amount each person gets, in dollars.

e) A phone plan costs \$25 a month plus \$2 for each gigabyte of data. Write an expression for the cost, in dollars, of a month using g gigabytes.

f) A rectangle is w cm wide. It is 3 times as long as it is wide. Write an expression for its perimeter.

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C Equations from words ●●○

Remember: An equation has an equals sign. Write the equation first, then solve it.

3. Write an equation for each problem, then solve it to find the number.

a) I think of a number and add 7. The answer is 20.

b) I think of a number, double it and subtract 3. The answer is 15.

c) A number divided by 5 is 6.

d) Four times a number, plus 5, equals 33.

e) When 7 is subtracted from 4 times a number, the answer is 25.

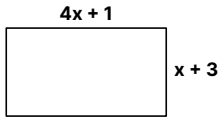
f) Half of a number, plus 5, is 11.

D Expressions for perimeter and area ●●●

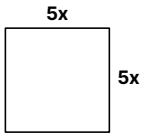
Remember: Perimeter: add all the sides. Area of a rectangle: length × width. Simplify your answer.

4. Write an expression for the perimeter of each shape.

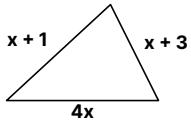
a)



b)

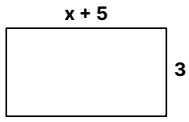


c)

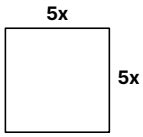


5. Write an expression for the area of each shape.

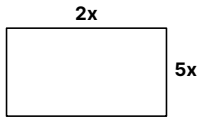
a)



b)



c)



E Solving problems ●●●

6. Write an equation for each problem, then solve it.

a) A rectangle is x cm wide and $(x + 6)$ cm long. Its perimeter is 36 cm. Find its width.

b) An adult ticket costs \$15 and a child ticket costs x dollars. Two adults and 4 children pay \$62 altogether. Find the cost of a child ticket.

c) Ana has x dollars. Ben has twice as much as Ana. Together they have \$72. How much does Ana have?

d) Two consecutive whole numbers add to 57. Find the smaller number.

★ Extension

7. n is a whole number. Write an expression for the sum of n and the next two whole numbers after it. Simplify your answer.

8. A square has sides of s cm. Each side is made 3 cm longer. Write an expression for the perimeter of the new square. Simplify your answer.

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